A Smartwatch Moves: The Impact of Walking Speed and Trajectories on Micro Visualizations Reading on a Smartwatch

Fairouz Grioui, Lijie Yao, Petra Isenberg, Tanja Blascheck

***Abstract*—We describe the impact of smartwatch wearers' motion on how effectively wearers can read micro visualizations, how challenging the task feels, and how these effects relate to smartwatch orientation. Visual information on smartwatches is often consumed in dynamic contexts that involve motion such as walking or running. Motion can introduce perceptual challenges due to the way people position their smartwatch, increasing the general task load, but these effects have not been systematically studied. Therefore, we measured smartwatch viewing postures and performance data while reading visualizations in two studies. These studies examined different walking trajectories and speeds with stationary baseline conditions. Our results show that walking and trajectory type altered smartwatch positioning and wrist orientation, with faster speeds increasing these effects. Walking, particularly at higher speeds, also reduced reading accuracy and sometimes increased response times, while participants also slowed their pace when reading micro visualizations. Based on these findings, we propose design considerations including additional visual cues and visualization size adaptations for different viewing postures to support glanceable micro visualizations in motion. The supplementary material for our studies can be found at https://osf.io/7pg2r.**

***Index Terms*—Micro Visualization, Mobile Visualization, Smartwatch**

I. Introduction

SMARTWATCHES and fitness bands are wearable devices worn on the wrist, which allow easy and quick access to the data the devices collect and display with just a twist of the wrist. Due to the compact form factor of smartwatch displays, visualizations are limited to a maximum size of roughly 5 cm when they occupy the entire screen. In practice, however, this area is often shared by multiple complications[1]—as is common on smartwatch faces [27]—further reducing the effective space available for visualization and small details. Beyond these spatial constraints, the mobile nature of smartwatch use in real-world contexts (e. g., walking and jogging in a park, or riding a bike) inherently involves additional challenges, where the device's distance and orientation to a viewer constantly change as the wearer moves. We know from prior work that the orientation or tilt of displays has an impact on how accurately people read visual variables and simple charts, either on static large displays [4], [55] or a small smartwatch screen [6]. However, to date, rare empirical measurements exist characterizing the range of movement of a worn smartwatch under real-world motion conditions. Moreover, although a previous study has investigated how well micro visualizations can be read on a static smartwatch face [6], we still know little about how the smartwatch's orientation, the viewer's motion, or both, may impact the perception of data from micro visualizations. On-the-go viewing of micro visualizations also has additional challenges, such as the presence of obstacles on the path, as well as added cognitive load of navigating, walking, and reading at the same time further impacting reading activities.

A deeper understanding of how such factors impact reading micro visualizations on the go can lead to future micro visualization designs that adapt (potentially automatically) to motion scenarios that are much unlike reading micro visualizations when stationary.

Therefore, it is important to conduct empirical studies on smartwatches while participants are in motion to generate this deeper understanding, and to elicit design implications. Yet, involving participant movement in studies proves to be challenging. Researchers need to balance between natural conditions during which participants move in everyday environments and the ability to measure how they read micro visualizations and position the smartwatch in a more constrained environment. In addition, the context, type, and speed of a person's activity impact movement characteristics, necessitating an established baseline and a community effort to run studies with multiple activities and in different contexts.

Prior work on reading phones in motion showed an impact on attention [24], [49] and motor skills [9], [16], potentially leading to decreased performance on a primary task (e. g., walking). Consequently, we posit that the best micro visualizations for smartwatches depend on the person's activity. Studying the differences between activities, however, requires a protocol to measure *how* a smartwatch moves *relative* to a viewer's eyes, to what extent viewing angles change when smartwatch wearers

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Manuscript received April 19, 2026; revised August 16, 2026.

[1]In horology, a complication is any graphical element presented on the watch face besides the time.

look at their devices while moving, and how the viewing distance between the smartwatch and the viewer changes during an activity. Currently, we know little about these concerns for any type of activity, and we have no baseline. As such, it is difficult to compare and explain why reading performance of micro visualizations might differ for certain activities or for different parameters of a similar activity, such as when walking at various speeds.

Our work tackles this underlying challenge of measuring the characteristics of smartwatch movement. As a baseline activity, we pick walking as a common activity and show how we can measure smartwatch motion when varying different walking parameters. This paper is a major extension of a prior IEEE VIS short paper [23] in which we reported on participants reading performance as well as on the perceived mental and physical load when performing two concurrent tasks (i. e., walking and reading micro visualizations on a smartwatch). In this extension, we now also assess how participants positioned their smartwatch, their viewing angle on the micro visualizations they read, and how the smartwatch moved in space during a walking activity at different speeds and trajectories.

We also provide and discuss our measurement protocol to help compare the results of future studies on reading micro visualizations in motion with similar or even different activities. Our complete work now creates a basis for comparative studies of micro visualization reading on smartwatches under motion.

II. Related Work

Previous research on mobile and wearable devices in visualization and human-computer interaction has explored the design of visual data representations [2], [5], [11], [26]–[28] and how to better display and interact with micro visualizations on a smartwatch [41]–[44]. Apart from the work from Islam et al. [29], which reflected on how motion characteristics might affect perception, design, and the experience of micro visualizations displayed on smartwatches, other works mentioned above mainly focused on the representations used on the display area of smartwatches. A real usage context is missing, in which smartwatch wearers need to read micro visualizations while walking, running, or, more generally, in motion. Such motion should not be ignored as it is not minimal—Gard and Childress [20] demonstrated the human body can move vertically from 3 cm to 8 cm (called the "vertical excursion") depending on different walking speeds. Also, recent work from Yao et al. [57] showed that stationary participants made more errors when reading from charts under higher moving speeds and irregular movement trajectories. Nevertheless, it is still unclear how such motion characteristics can affect reading accuracy and posture when both the viewer and the visualization are in motion. We focus on mobile and wearable devices, specifically smartwatches, to fill this gap. Next, we introduce related work about micro visualizations on smartwatches, as well as the study of smartwatches and smartphones in motion.

A. Micro Visualizations on Smartwatches

Previous visualization research has contributed novel, space-efficient *visualization designs* for displaying complex micro visualizations, including large time series [11], multiple interlinked time series [41], and line charts with compressed axes [44]. Researchers also looked at people's ability to (quickly) *interpret* simple micro visualizations (e. g., donut charts, bar charts [5], [6], [22]) and other conventional micro visualization techniques (e. g., hypnograms to represent sleep data [26], and Time Spirals [51] to self-track back pain) when scaled down to accommodate the small screen of a smartwatch. This past work has explored several avenues for using micro visualizations on small screens: using simple, low-dimensional visualizations [5], [6], creating techniques to adapt visualizations to the small screen [44], asking participants which visualizations they would prefer on the small screen [26], or interacting with smartwatch visualizations [42], [46]. Other prior work explored extending the display space to smartwatch straps [32] for displaying data, including line charts and maps. Most previous studies on micro visualizations for wrist-worn devices were conducted using a stationary setup. Studying micro visualization reading in stationary settings has several benefits: it allows to control the viewing distance and angle variability, and as such reduces measurement noise, compared to settings with motion. In our work, we expand on these past efforts by proposing a study and data analysis protocol that helps to capture the nuances of motion together with quantitative measures of task performance and to compare and explain how results might change in motion.

B. Smartwatch Usage in Motion

Most prior work studying smartwatches in motion used treadmills to put participants in motion. One set of studies in this context focused on **interaction techniques**. For example, Neshati et al. [42] considered three interaction techniques to reduce screen occlusion when interacting with graphs on a smartwatch while standing and walking on a treadmill (with speed adjusted to participants' preferences). Their results showed no difference in response time between the standing and walking conditions per interaction technique. Turner et al. [53] compared two typing methods—trace and tap—using a full QWERTY keyboard on a smartwatch while standing and walking on a treadmill. The researchers found that walking affected typing speed and accuracy. Mo et al. [37] studied the impact of interacting with a smartwatch by tapping, swiping, and wrist flicking on the wearer's motion and gait patterns while standing and walking. The researchers examined four walking speeds: strolling, regular walking, rushing, and jogging. The results showed that the three interaction techniques impacted the gait symmetry and decreased the step length while walking. The effectiveness of tapping was significantly affected when the walking speed increased.

Other studies explored the use of **specific micro visualizations** on the go [30], [31], [48]. These studies consis-

tently demonstrated that graphical visualizations outperformed text-based display on smartwatch screen during motion; achieving shorter reading times, more accuracy, and greater subjective satisfactions. Schiewe et al. [48] evaluated textual feedback with two micro visualizations—showing heel and forefoot strikes—on a smartwatch to guide runners in forefoot training on a treadmill. Kashanj et al. [31], demonstrated in a real-world outdoor running study that visualizations outperformed text-based displays, enabling runners to complete information retrieval tasks (1.5 to 8 times faster) with a strong subjective preference for visual formats. Complementing this finding, Jin and Deng [30] conducted a more granular investigation across three motion conditions: static, low- and high-intensity movement on a treadmill, examining not only presentation format but also animation style and color mode. Their results revealed that non-animated presentations yield better cognitive efficiency and that dark mode consistently reduced visual fatigue during exercise.

Grioui and Blascheck [22] simulated a tennis-like game in VR, during which participants had to move laterally, and evaluated the efficiency of three heart-rate visualizations displayed on a virtual smartwatch to monitor a fitness goal. However, neither of the aforementioned studies reported on the effect of movement on reading the visualizations on the smartwatch. In other more recent works investigating reading simple visualizations [23], [56] and comparing the effectiveness of pre-attentive variables [21] on smartwatches while walking, the results showed that the reading performance, in particular of details (i.e., percentage estimation) was affected by motion.

In an attempt to understand how viewers concretely position their smartwatch while reading textual information, Blascheck et al. [5], [7] measured the orientation of the smartwatch and the viewing distance between the smartwatch and the viewer using a motion-tracking system with stationary postures (i. e., sitting and standing). They found that, on average, participants positioned their smartwatch at a distance of 28 cm and tilted their wrist by 40° when reading data on a smartwatch.

In our paper, we focus on motion scenarios with movement of a viewer and a micro visualization. We do not discuss scenarios in which a viewer and a micro visualization move together (e. g., a car driver reading a fixed heads-up display [38] or a touch screen in a simulated vibration environment [10]).

In summary, the scope of implications of smartwatches in prior research examining reading micro visualizations in motion was noticeably restricted. Our work provides measured, empirical evidence on how motion affects reading performance and posture, walking speed, and the extent to which reading micro visualizations in motion impacts the task's mental and physical load.

C. Smartphone Usage in Motion

Although our work does not focus on smartphones, a related body of work explored how people read from smartphones on the go and how motion factors affect **reading posture**, **reading accuracy**, **cognitive load**, and **moving speed**, which are close to our research questions. Reading on the go has been studied under the lens of dual-task performance [9], [16], [35] to assess the interdependence between a primary engaging task and a secondary task. For instance, when a pedestrian has to handle multiple tasks simultaneously (e. g., reading a notification while walking), their cognitive abilities can be divided, causing one task to take precedence over the other. As a result, a decline in attention and performance can occur, increasing the probability of accidents [9], [16], [24]. Related to our context of reading micro visualizations, prior work examined reading text on a mobile phone while walking [39], [49]. Mustonen et al. [39] compared text legibility with regular reading and pseudo-text search tasks in four walking conditions. They found that increasing walking speed negatively affected reading speed. Schildbach and Rukzio [49] found a decrease in performance and an increase in cognitive load in tasks that involved target selection and reading on a mobile phone while walking. Increasing the text size improved target selection performance but not reading performance, due to the need for scrolling. Previous work investigated texting while walking and found that the activity affected the walking style by decreasing the gait speed [1], [35], [45]. Also, other studies examined how different viewing conditions impacted the neck and head posture when interacting with mobile devices. Young et al. [58] explored flexion angles, viewing distance, and angle variation with four sitting conditions in one case (two placement conditions: on the lap and the table; two support conditions: handheld). Their results showed that head and neck flexion angles with the tablet condition were greater than those in previous measures with desktops. The authors recommended placing tablets on high surfaces with steep tilt angles to reduce the risk of discomfort when using them. Besides, previous studies have shown that viewers' head motion is less smooth and regular at higher walking speeds and can slow viewers' walking speed [47].

Smartwatches present major differences from mobile devices in terms of their form factor (i. e., the screen size and shape) and the way they are positioned for reading the screen—they are worn around the wrist instead of being held. This induces additional constraints related to the anatomy and range of motion of the arm, which, to our knowledge, has not been studied before. In this paper, we report on posture measurements while reading visualizations on a smartwatch, in addition to reading performance and tasks' subjective mental and physical loads under mobility and stationary conditions.

III. Reading Micro Visualizations in Motion

Various factors influence how humans walk and perform tasks while in motion. Factors such as the complexity of the walking trajectory (including obstacles on a path) and the walking speed can influence the task of reading relevant information from a smartwatch and the way people

walk while reading. We, therefore, conducted two studies to assess how motion affects viewers' reading performance and posture while walking. We varied the walking trajectories in Study 1 and the walking speed in Study 2 to address the following research questions:

RQ1. What is the influence of motion on the reading performance of micro visualizations while walking?

RQ2. How does motion affect people's posture while reading micro visualizations on a smartwatch?

RQ3. What is the influence of different walking speeds and trajectories on the subjective mental and physical load of the performed task?

RQ4. How does reading micro visualizations on a smartwatch affect walking speed?

A. Task and Stimulus

In this section, we describe our design choices for the displayed micro visualizations and the reading task.

1) Visualization Design Choices: A watch face is the primary interface on a smartwatch when activated. Typically, it shows about five complications, including a collection of glanceable information [27]. We designed a canonical watch-face stimulus showing date, time, and weather information at the top and a set of personal data at the center and bottom of the screen (Fig. 1). Rectilinear and radial progress charts (e. g., bar, and donut charts) are commonly used visual representations on watch faces [6], with radial representations being the most dominant. They often depict fitness data progress towards a goal, for example, for calories, distance, and steps, as well as exact quantities relative to a chosen maximum for heart rate or stress levels. At the center of the screen, we rendered three juxtaposed radial progress charts showing three different fitness data. The charts were uniformly scaled to a diameter of 0.9 cm each to ensure they fit within the smartwatch display area while maintaining sufficient inter-element spacing. A distinct color was associated to each progress chart: **calories burned** (cb), **step count** (sc), and **distance walked** (dw). The unfilled part of the bar was colored in gray. Inside each chart, we displayed text showing the current progress and target values. The values described a low to average activity level (i. e., 6,000 steps/day). Participants had to read these charts during our studies. We also displayed heart rate information as a curved bar chart (164) to resemble a canonical watch face. Its color depicted the Heart Rate Zone (HRZ) information (light blue: low, orange: moderate, red: high HRZ).

2) Task Design: Previous studies [2], [13] showed that people are interested in estimating the progress towards a goal when tracking their fitness data on the go. Therefore, our task asked: *which fitness data has the highest progress towards a goal, and how much is this progress in percent?* Specifically, we asked participants to compare the proportions of the three radial charts, indicate the fitness data with the highest percentage, and estimate the percentage of this chart. We created nine distinct stimuli with different percentages. Each fitness data type had the highest value in three stimuli (Fig. 1). All proportion percentages were multiples of 5, and we informed participants of this. The difference between the radial progress chart with the highest progress and the two others was always 5% and 10%. We used the same stimuli in both studies but randomized their order between conditions.



Fig. 1: Examples of watch-face stimuli with three radial progress charts: **calories burned (cb)**, **step count (sc)**, and **distance walked (dw)**. The visualized progress percentage (the highest highlighted in color) on each stimulus is: **cb: 75%**, dw: 70%, sc: 65% (left), **sc: 65%**, dw: 60%, cb: 55% (middle), and **dw: 50%**, cb: 45%, sc: 40% (right).



Fig. 2: The head marker used in Study 1 (left), in Study 2 (middle), and the smartwatch marker (right) used to track the head and the smartwatch position and orientation.

B. Apparatus

The studies were conducted in two different labs. The first had a walking space of 3×5 m. In the second, we used an electric Christopeit Sport TM2 Pro treadmill, whose speed was set manually by the researcher conducting the study at the beginning of each walking condition. We displayed the stimuli on a Fossil Gen 6 smartwatch with a screen size of 44 mm and a resolution of 326 ppi. We asked participants to wear reflective markers that were registered with a 3D motion tracking system (Sect. III-G). We placed one around the left wrist (under the smartwatch) and the second on a headband around the forehead of the participants (Fig. 2). We used different wrist and head markers for each study because the two studies were conducted in different labs with different tracking systems. However, we ensured that the markers were configured the same way. The only difference between the two markers

TABLE I: Participants' demographic data.

Study	Age (yrs)	Height (cm)	Arm-length (cm)
Study 1	M=28.4 SD=4.5	M=172.5 SD=9.4	M=54.7 SD=4
Study 2	M=26.8 SD=4.7	M=176.4 SD=10.7	M=55.4 SD=4.6

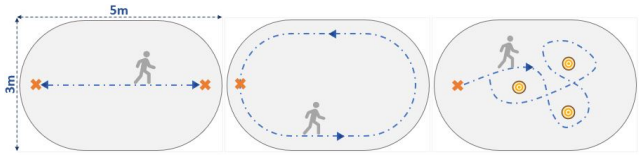


Fig. 3: Illustrations of the three walking trajectories from Study 1 (left to right): line, circular, and infinity.

was that the head marker in Study 1 was placed on a vertical stick at a distance of 12 cm from the center of the forehead. The marker in Study 2 was positioned directly at the center of the forehead. We accounted for this difference when computing the posture measurements.

C. Participants

We recruited 18 participants: (4 women, 14 men) in Study 1, and 24 participants: (9 women, 15 men) in Study 2; for details see Table I. Eight participants from Study 1 and twelve from Study 2 reported owning a smartwatch or fitness band. Four participants from Study 1 and seven from Study 2 used the device daily. All participants were right-handed, and apart from one participant in Study 1, all reported wearing watches on their left wrist. All participants confirmed that they had good eyesight, including 10 in Study 1 and 8 in Study 2, who wore glasses.

D. Study Procedure and Design

After signing the consent form, we explained the study task to the participants. Then, we tested their ability to read radial chart proportions, which all participants did correctly. Each participant performed all conditions and started with a training phase. The order of execution of the conditions was counterbalanced. Participants were shown nine stimuli per condition, each for 12 s max. Both studies were approved by our university ethics committee.

E. Three Trajectories Conditions

For Study 1, we designed three walking trajectories (Fig. 3) to mimic different real-world walking situations.

⟷ **Line:** Straight trajectory, with known start/end points. Participants took nine laps, one per stimulus.

Circular: A continuous, oval-shaped trajectory.

Infinity: Participants walked continuously around three cones, creating an infinity-like trajectory.

During the ⟷ line trajectory condition, the walked distance was quite short. Therefore, we asked participants to press the smartwatch screen at the beginning of each lap to trigger the visualization display and ensure that participants read the micro visualizations while walking. In contrast, for both the circular and the infinity trajectories, participants pressed the smartwatch screen only once at the beginning, when they started walking. Every 12 s, a new visualization was displayed on the screen, and participants were notified with a vibration from the smartwatch to look at the stimulus (Fig. 4). We did not prompt participants to perform the task as fast as possible. However, they were informed about the 12 s delay.

F. Walking Speed and Stationary Conditions

In Study 2, we tested three walking speeds and two stationary conditions. The choice of speeds was guided by the need to provide increasingly fast but distinguishable conditions feasible for a heterogeneous participant sample of varying body metrics e. g., step lengths and varying fitness levels.

2 **Slow Walking Speed.** In the first study, participants walked at a normal pace with an average speed of 2.5 km/h. Therefore, we set the slow walking speed on the treadmill to a reduced speed of 2 km/h to capture low-intensity movements.

4 **Moderate Walking Speed.** We operated the treadmill at a speed of 4 km/h (≈ preferred walking speed [40]), to simulate a moderate walking speed.

6 **Slow Jogging Speed.** We set a fast walking to slow jogging speed of 6 km/h to approach the upper of brisk walking without requiring participants to transition into running.

Stationary Conditions. Participants performed the task while sitting on a chair and while standing.

G. Measurements

The recorded measurements helped to identify phases during which participants walked and viewed the smartwatch and phases during which they were solely walking. As shown in Fig. 5, we selected the smartwatch viewing phases manually based on a) indications that the arm was held closer to the head (Line of Sight (LOS) distance) and b) that the watch face was oriented toward the participants' eyes (Alpha-Floor angle). We collected the measurements outlined next, to analyze participants' task performance, walking, and reading behavior during the micro visualizations reading phases.

Reading Measurements

To avoid interrupting participants' movement while reading the visualizations, they were instructed to verbalize their answer by naming the data type with the highest progress, followed by its estimated percentage.

Response Time. We measured the time participants took to verbalize their response for each stimulus when their arm was in a stable reading position. Two authors of the paper extracted response time phases using a precise frame-based video annotation tool. The tool allowed a coding with a frame-by-frame playback at one-tenth of the original video speed. In Study 2, we only analyzed 75% of the videos because, for two participants the video records were missing a subset of trials (ranging from one to four trials across the four conditions). For another four participants, entire condition trials were missing (3× sitting and 1× for each of the four other conditions). The data loss was not balanced between the study conditions.

Reading Accuracy. During the study, one researcher noted participants' spoken answers in a separate spreadsheet, then collected whether or not participants

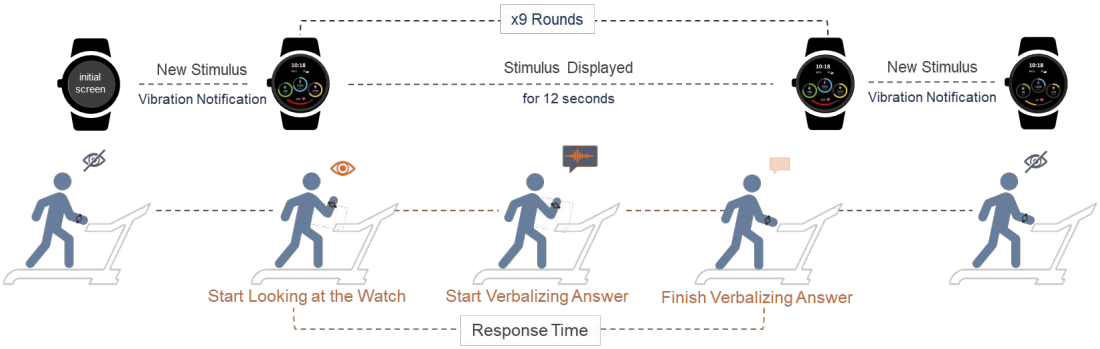


Fig. 4: Illustration of the micro visualizations reading task that participants performed in the two studies.

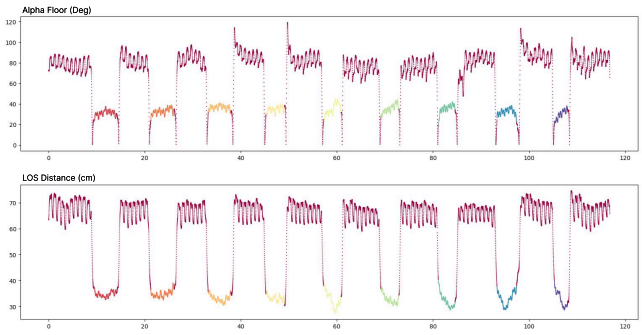


Fig. 5: An example of the Alpha-Floor angle and LOS distance measurements over time for one participant. The nine reading phases are highlighted with different colors.

correctly identified the fitness data with the highest progress (*fitness data comparison accuracy*). For all correct answers of data types, we additionally analyzed the difference between the provided and the correct percentages (*proportion estimation accuracy*). An answer was considered as correct only if it exactly matched the correct answer.

Posture Measurements

We used a Vicon tracking system with 16 (Study 1) and 8 (Study 2) infrared cameras to record the 3-dimensional position and a quaternion matrix to describe the global angles of the head and wrist marker.

In the following, we report on the LOS distance between the forehead and the smartwatch marker centers, in addition to the three-dimensional rotation of the smartwatch, according to a fixed global reference (i. e., the floor normal and the head forward x-axis (Fig. 6)).

LOS Distance. The Euclidean distance between the center of the smartwatch and the center of the forehead.

LOS Distance Standard Deviation. We measured the mean standard deviation of the LOS distance to gauge how much the smartwatch viewing distance deviated from the mean LOS distance during the reading phases. This metric gives an indication about how much movement there was of the smartwatch while a participant was reading a visualization.

Alpha-Floor Angle (α-Floor). The Alpha-Floor angle is the rotation of the smartwatch around its x-axis, in reference to the floor normal (Fig. 6 (a)). It is calculated as the angle between the projection of the floor normal v(0,0,1) onto the y-z plane of the smartwatch and the smartwatch's z-axis.

Beta-Floor Angle (β-Floor). Represents the rotation around the smartwatch's y-axis with reference to the floor normal (Fig. 6 (b)). It is calculated as the angle between the projection of the floor normal v(0,0,1) onto the x-z plane of the smartwatch and the smartwatch's y-axis.

Gamma-Head Angle (γ-Head). Represents a rotation around the smartwatch's z-axis in reference to the participant's forward axis (Fig. 6 (c)). It is calculated as the angle between the projection of the head's x-axis (i. e., the head forward axis) on the x-y plane of the smartwatch and the smartwatch's y-axis.

Walking Speed

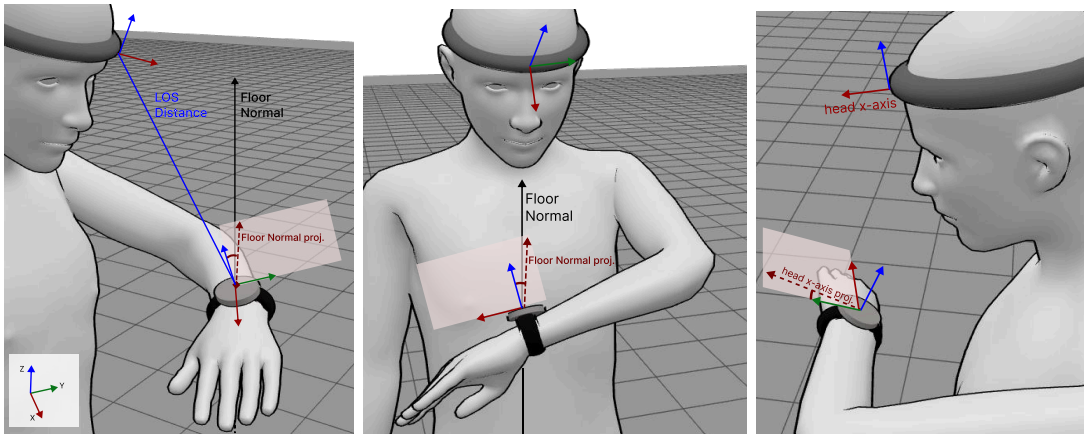
During Study 1, we measured participants' walking speed. The Vicon tracking system measures data according to a specific frame rate, in our case 120 Hz. We measured the walking speed using the recorded frames with the following formula: $speed = distance2frames/durationPerFrame$, where $distance2frames$ represents the distance in meters between two head positions for two consecutive frames, and the $durationPerFrame(s) = 1/frameRate$.

Subjective Measurements

For each condition, participants used a 7-point rating scale to describe their perceived task difficulty (1: not at all difficult, 7: very difficult), their confidence with their answers (1: not at all confident, 7: very confident), and the subjective mental and physical load of the task (1: not at all demanding, 7: very demanding).

IV. Results

We report sample means of the reading performance for 486 trials (18 participants × 9 stimuli × 3 conditions) from Study 1 and 1080 trials (24 participants × 9 stimuli × 5 conditions) from Study 2 using 95% confidence interval (CI) estimation [3], [14], [17], [18] using BCa bootstrapping (10,000 bootstrap iterations). In addition, we report the differences between means to compare the three conditions. We adjusted the CIs of mean differences for multiple



(a) α-Floor (b) β-Floor (c) γ-Head

Fig. 6: Three illustrations showing the posture measurements with the wrists three rotation angles: (a) rotation around the smartwatch's x-axis relative to floor normal (α-Floor), (b) rotation around the smartwatch's y-axis relative to floor normal (β-Floor), (c) rotation around the smartwatch's z-axis relative to head forward axis (γ-Head).

comparisons using Bonferroni corrections [25] to ensure consistency between the two studies. When reading a CI of mean differences, we interpret a non-overlap of the CIs with 0 as evidence of a difference, which correspond to significant results in traditional p-value tests. We use the qualitative terms strong evidence, evidence, weak evidence, and trend descriptively to summarize the location and precision of the bootstrap confidence intervals, ordered from the furthest to the closest (or marginally overlapping) relative to the zero line, respectively. In this section, we present our two studies' results, including the reading measurements, the posture measurements, the influence of reading micro visualizations on walking speed, and the subjective measurements. All measurements were calculated only for the micro visualizations reading phases.

A. Reading Measurements

We compare reading performance within the two studies' walking conditions, as well as between walking and stationary conditions in Study 2 to assess the influence of multitasking and motion on reading micro visualizations.

Response Time. On average participants read the micro visualizations in 5.7 s (and : 5.5 s and ⟷: 6.0 s) in Study 1 (Table II, Row 1). The analysis of pairwise differences in Study 1 shows no evidence of a difference between the conditions. The average response time in Study 2 is 3.9 s (: 3.65, 4 : 3.71 s, 6 : 3.91 s, : 3.95 s, and 2: 4.41 s; Table III, Row 1). The mean difference results shows trends that the mean response time with the 2 km/h condition is longer than the 4 km/h and the standing conditions.

Reading Accuracy. Participants' answers about the fitness data with the highest progress were, on average, 94.8% accurate with the three trajectories (: 93.8%, : 94.4%, and ⟷: 96.3%; Table II, Row 2). Participants also read the percentages on average with a correctness of 72.2% (: 70.4%, : 71.6%, and ⟷: 74.7%; Table II, Row 3). The analysis of pairwise differences in Study 1 shows no evidence of a difference between the conditions. In Study 2, the average answers accuracy about the fitness data was 94.7% with the three walking speeds (6: 93.5%, 4: 94.9%, and 2: 95.8%), and 97% with the stationary conditions (: 95.4% and : 98.6%; Table III, Row 2). The percentages were estimated with an accuracy of 78.2% with the three walking speeds (6: 74.5%, 4: 79.2%, and 2: 81%), and 82.2% with the stationary conditions (: 80.6% and : 83.8%; Table III, Row 3). Participants' answers were more accurate with the standing condition compared to the three walking speeds, particularly the 6 km/h condition. For the proportion estimation answers, the pairwise differences show weak evidence that participants were less accurate with the 6 km/h condition in comparison to standing. In addition, the results showed trends that the proportion reading was more accurate with the 2 km/h condition and while sitting compared to the 6 km/h condition.

B. Posture Measurements

One of our core contributions is results related to how people position their smartwatches while walking. In the following, we report the posture measurements from the two studies, highlighting variations between conditions.

LOS Distance. In Study 1, on average, participants held the smartwatch at a distance of 27.9 cm (: 27.5 cm, : 28 cm, and ⟷: 28.2 cm) from their eyes while walking. The difference between the three walking trajectories is less than 1 cm. Hence, the results show no evidence of pairwise differences in LOS distance between walking trajectories (Table IV, Row 1). In Study 2, on average, participants read the visualizations from a distance of 30.1 cm in the stationary conditions (: 29.8 cm, and : 30.4 cm) and 31.2 cm while walking (4: 31 cm, 2: 31.3 cm, and 6: 31.4 cm). Our results show weak evidence of a difference between the sitting condition and the three walking conditions, in addition to trends between the standing condition and all the other conditions except the 4 km/h condition (Table V, Row 1).

TABLE II: Means (top) and pairwise difference (bottom) of reading performance for Study 1: response time (Row 1), fitness data comparison accuracy (Row 2), proportion estimation accuracy (Row 3). Error bars represent 95% Bootstrap CIs in black, adjusted for three pairwise comparisons with Bonferroni correction in red. ⟷ line, ⟲ circular, 🤞 infinity trajectory .

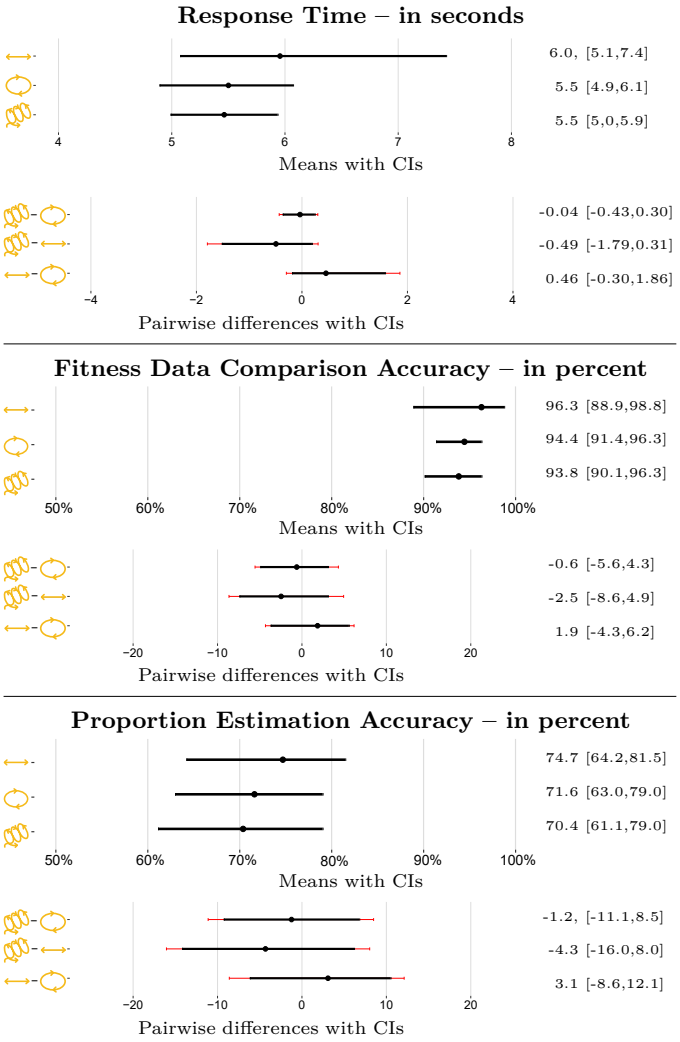
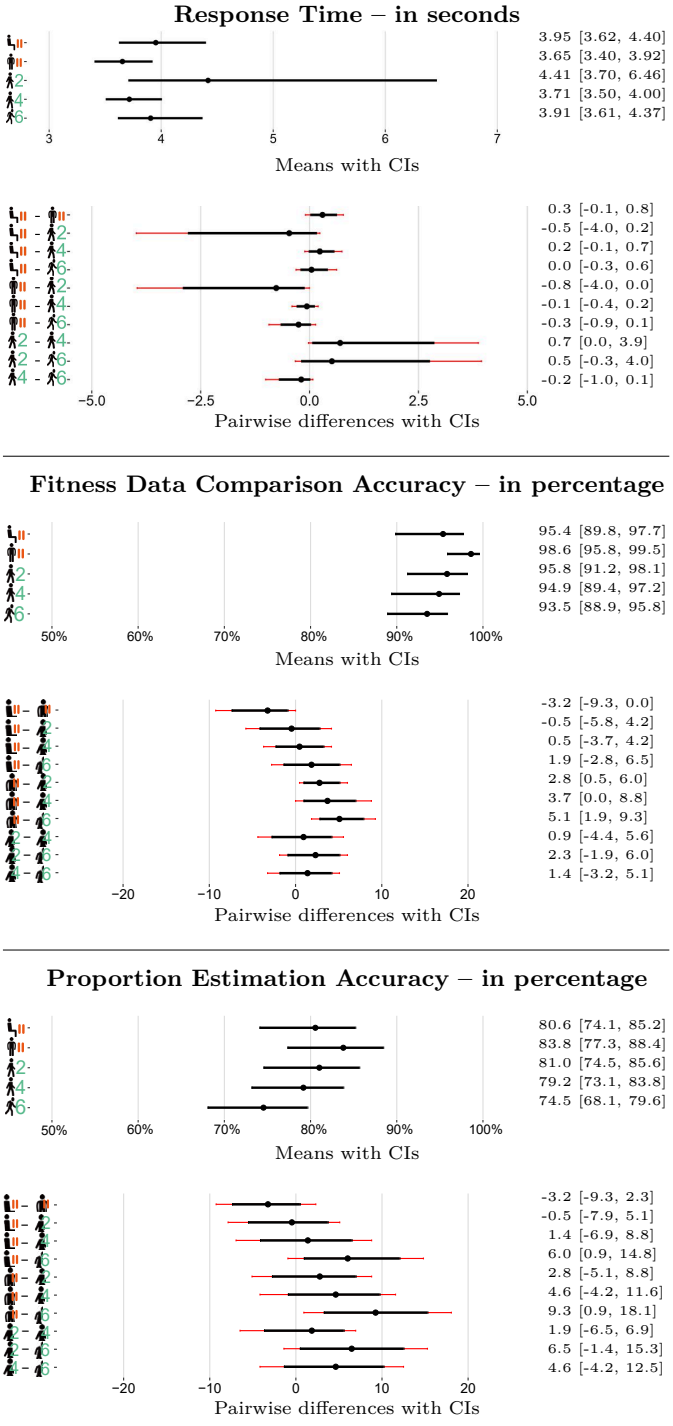


TABLE III: Means (top) and pairwise differences (bottom) of reading performance for Study 2: response time (Row 1), fitness data comparison accuracy (Row 2), proportion estimation accuracy (Row 3). Error bars represent 95% Bootstrap CIs in black, adjusted for ten pairwise comparisons with Bonferroni correction in red. 🧍 Standing, 🪑 sitting, 🚶2 km/h, 🚶4 km/h, 🚶6 km/h .



LOS Distance Standard Deviation (SD). In Study 1, the average LOS distance SD was 1.8 cm with the three walking trajectories (⟲ and 🤞: 1.7 cm, and ⟷: 2.1 cm). The pairwise differences show a trend that the LOS distance SD was higher with the ⟷ line trajectory than the ⟲ circular and 🤞 infinity trajectories (Table IV, Row 2). In Study 2, the average LOS distance SD was 1.3 cm with the three walking conditions (🚶2 and 🚶4: 1.2 cm, and 🚶6: 1.4 cm) and less than 1 cm with the stationary conditions (🪑: 0.7 cm, and 🧍: 0.9 cm). Our results show evidence of a difference between all stationary and walking conditions, in addition to the 🚶4 km/h and 🚶6 km/h conditions. The results also show trends that the LOS distance varied more with the 🚶6 km/h condition than the 🚶2 km/h condition (Table V, Row 2).

Alpha-Floor angle (α-Floor). The smaller the Alpha-Floor angle, the more parallel to the ground the smartwatch was positioned. Higher Alpha-Floor angles mean that participants rotated their wrists more toward themselves (around the watch's x-axis). On average, for all three walking trajectories, participants looked at the

TABLE IV: Means (top) and pairwise differences (bottom) for Study 1: LOS distance (Row 1) and LOS distance standard deviation (Row 2). Error bars represent 95% Bootstrap CIs in black, adjusted for three pairwise comparisons with Bonferroni correction in red. ⟷ line, ◯ circular, ∞ infinity trajectory .

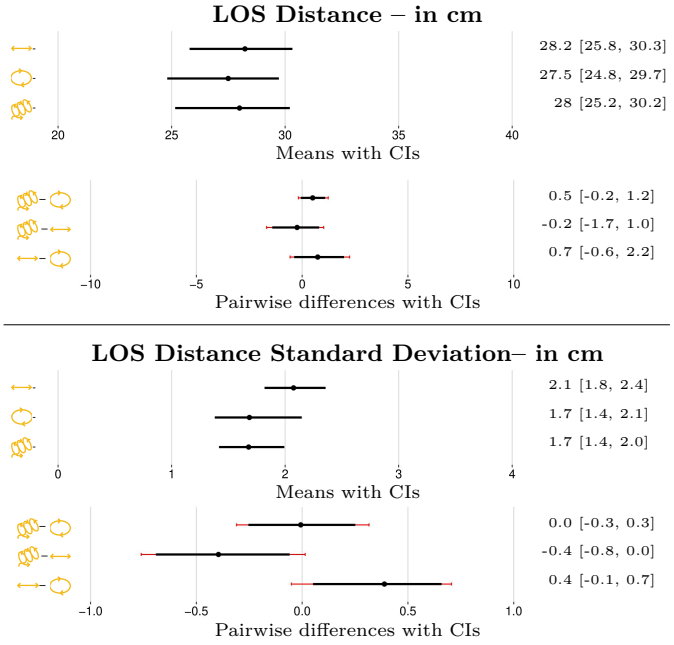
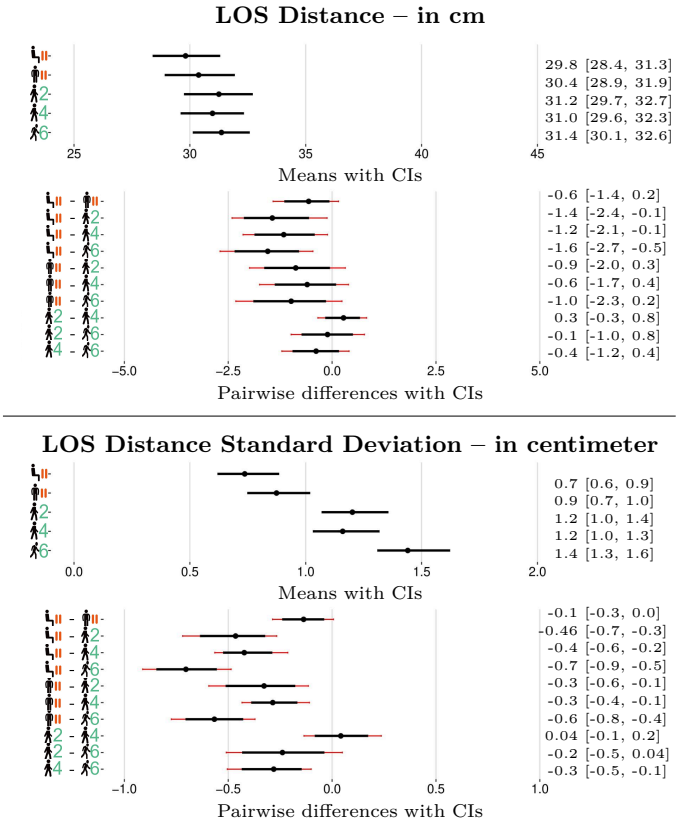


TABLE V: Means (top) and pairwise differences (bottom) for Study 2: LOS distance (Row 1) and LOS distance standard deviation. (Row 2). Error bars represent 95% Bootstrap CIs in black, adjusted for ten pairwise comparisons with Bonferroni correction in red. Standing, sitting, 2 km/h, 4 km/h, 6 km/h .



smartwatch with an Alpha-Floor angle of 27.4° (⟷: 25° (SD: 3.3°), ∞: 28.5° (SD: 3.5°), and ◯: 28.7° (SD: 3.2°)). A comparison of pairwise differences of the Alpha-Floor angle show evidence of a difference between the ⟷ line compared to the ◯ circular, and ∞ infinity trajectories (Table VI, Row 1). In Study 2, the average Alpha-Floor angle was 32.9° while stationary (: 30.5° (SD: 1.4°) and : 35.2° (SD: 1.1°)) and 27.7° while walking (2 : 25.8°, (SD: 2.1°) 4 : 28.2° (SD: 2.3°), and 6 : 29.1° (SD: 4°)). Our results show evidence of a difference between the sitting condition and all the other conditions. Strong evidence (5°–10°) was found with the 2 km/h and 4 km/h conditions. Also, we found a significant difference in the Alpha-Floor angle between the standing and 6 km/h condition. The results show trends that the Alpha-Floor angle was lower with the 2 km/h condition compared to the two other walking speeds (Table VII, Row 1).

Beta-Floor angle (β-Floor). The smaller the Beta-Floor angle, the more parallel to the ground the smartwatch was positioned. Higher Beta-Floor angles indicate that participants moved their elbows up or down relative to the wrist, resulting in rotation around the watch's y-axis. On average, participants viewed the smartwatch with a β-Floor of 13.1° in Study 1 (⟷: 12° (SD: 3°), ∞: 12.8° (SD: 3.3°), and ◯: 14.5° (SD: 2.9°)). A comparison of pairwise differences in β-Floor between the three walking trajectories shows evidence of a difference between the ◯ circular trajectory and the ⟷ line and ∞ infinity trajectories (Table VI, Row 2). In Study 2 participants viewed the smartwatch with an average β-Floor of 5.5° (2 : 4.9° (SD: 1.9°), 6 : 5.1° (SD: 2.2°), 4 : 5.3° (SD: 2°), : 6° (SD: 1.1°), and : 6.2° (SD: 1°)). The results show trends that the Beta-Floor angle was lower with the 2 km/h condition compared to the two stationary conditions (Table VII, Row 2).

Gamma-Head angle (γ-Head). The higher γ-Head, the more the smartwatch was rotated relative to the participant's forward direction. For example, a smartwatch lying flat on a table in front of a person would produce an angle of 90° when turned toward the viewer by 90° γ-Head will be 0°. On average, for all three walking trajectories, participants looked at the smartwatch with a γ-Head of 26.8° (◯: 24.2° (SD: 3.7°), ∞: 27.7° (SD: 4.3°), and ⟷: 28.6° (SD: 4.8°)). Similarly to the Beta-Floor angle, the results show evidence of a difference between the ◯ circular trajectory and the two other trajectories (Table VI, Row 3). The average γ-Head in Study 2 was 12° while stationary (: 11.8° (SD: 1.9°) and : 12.1° (SD: 1.9°)) and 13.8° while walking (6 : 12.9° (SD: 3.1°), 2 : 13.8° (SD: 2.5°), and 4 : 14.6° (SD: 2.5°)). A comparison of pairwise differences of γ-Head between the three walking speeds shows evidence of a difference between the

TABLE VI: Means (top) and pairwise differences (bottom) of the posture measurements for Study 1: Alpha-Floor (Row 1), Beta-Floor (Row 2), and Gamma-Head angle (Row 3). Error bars represent 95% Bootstrap confidence intervals (CIs) in black, adjusted for three pairwise comparisons with Bonferroni correction (in red). ⟷ line, ◯ circular, 🤜 infinity trajectory .

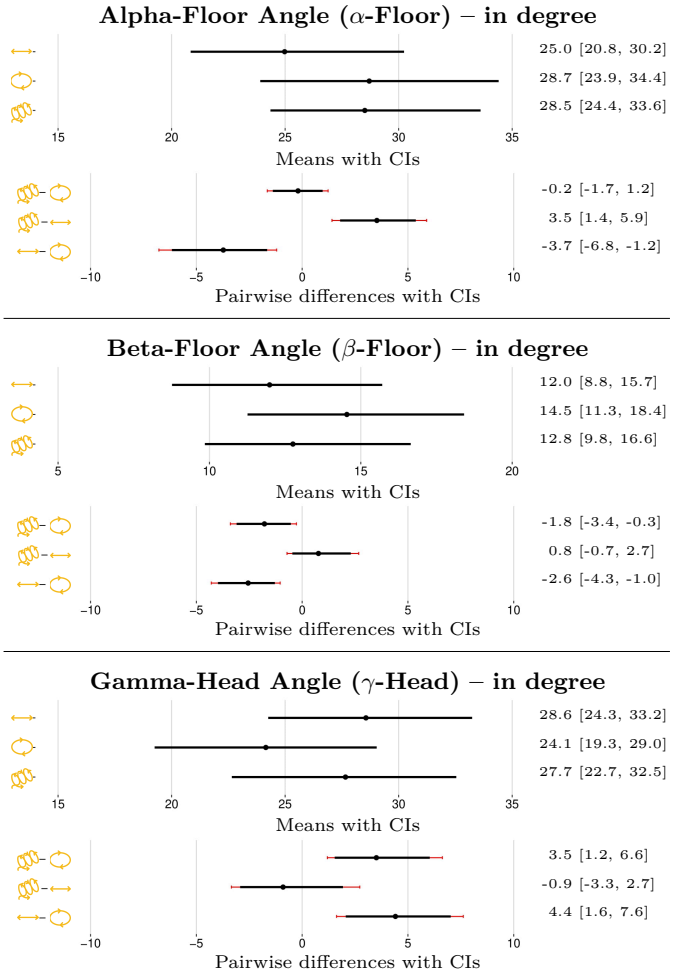
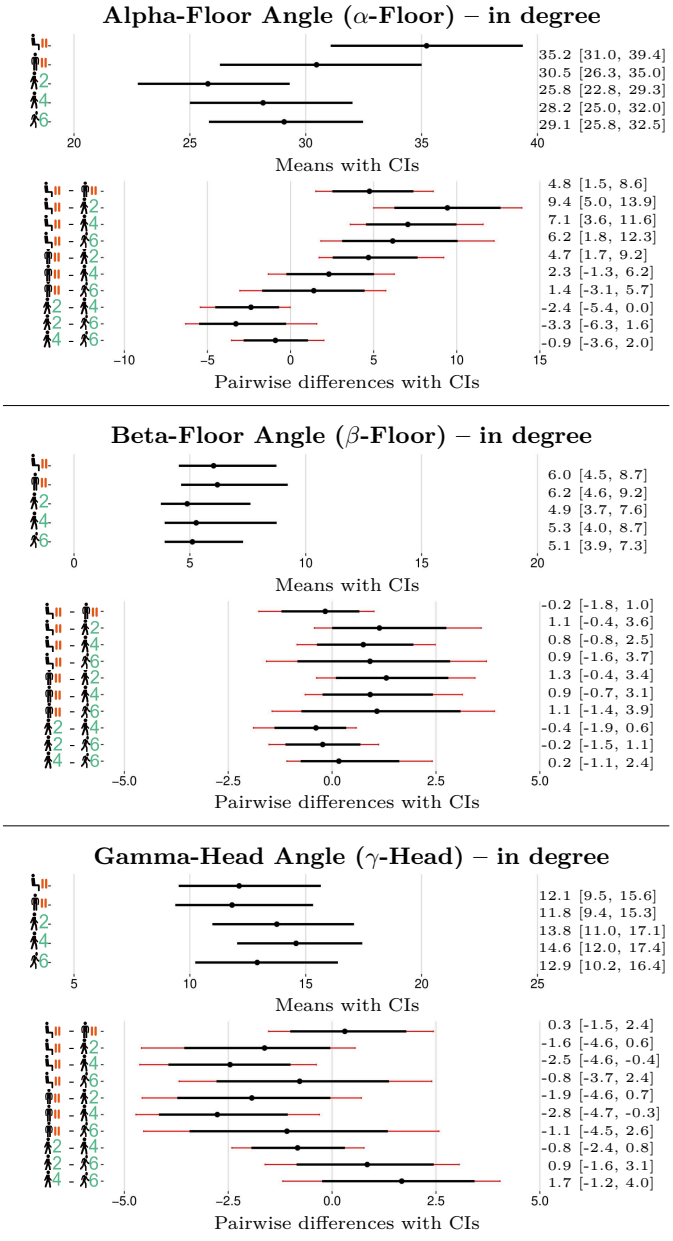


TABLE VII: Means (top) and pairwise differences (bottom) of the posture measurements for Study 2: Alpha-Floor angle (Row 1), Beta-Floor angle (Row 2), Gamma-Head angle (Row 3). Error bars represent 95% Bootstrap confidence intervals (CIs) in black, adjusted for ten pairwise comparisons with Bonferroni correction (in red). Standing, sitting, 2 km/h, 4 km/h, 6 km/h .



4 km/h condition and the two stationary conditions. The results also show that γ-Head was higher with the 2 km/h condition and the two stationary conditions (Table VII, Row 3).

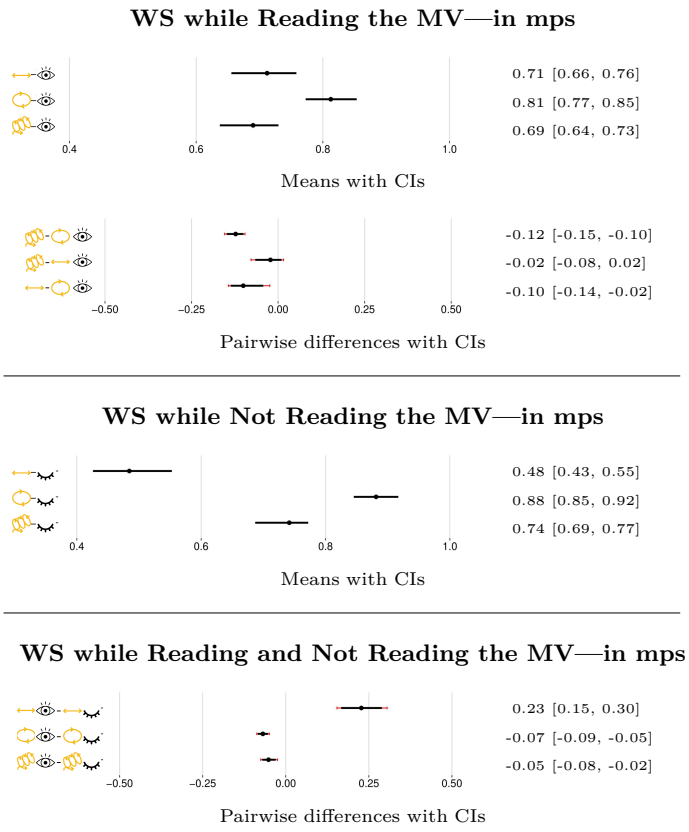
C. Walking Speed

In Study 1, the average walking speed while reading the micro visualizations was 0.7 m/s (= 2.5 km/h) (🤜 and ⟷: 0.7 m/s and ◯: 0.8 m/s). With the ⟷ line trajectory, participants read the micro visualizations when walking between two defined points. According to our observations, when they finished the task, participants were almost stationary or slowed down their walking speed because they were close to the end of the trajectory. For this reason, there was strong evidence for the ⟷ line trajectory that they walked faster while reading the micro visualizations compared to when they were not reading from the smartwatch (Table VIII, Row 3). In contrast, for the ◯ circular trajectory and 🤜 infinity trajectory, there is weak evidence of a difference in walking speed for a slower walking speed when reading the micro visualizations compared to when not reading from the smartwatch. Additionally, the pairwise comparison in walking speed while reading from the smartwatch indicate strong evidence of a difference that participants walked faster when performing the ◯ circular trajectory (Table VIII, Row 1).

D. Subjective Measurements

In the following, we report on participants' subjective evaluation (see supplemental material for more details).

TABLE VIII: Means and pairwise differences of walking speed (WS) for Study 1 while reading the micro visualizations (MV) on the smartwatch (Row 1). Means of walking speed while not reading from the smartwatch (Row 2, top) and pairwise differences in walking speed between reading and not reading the micro visualizations (Row 2, bottom) within each walking condition using 95% CIs and adjusted for three pairwise comparisons with Bonferroni corrections (in red). ⟷ line, ◯ circular, ∞ infinity trajectory, 👁: reading, 👁̸: not reading, mps: meter per second .



Confidence in Provided Answers. In Study 1, participants were fairly confident with their responses while walking the three trajectories (median: 5/7). Likewise, their confidence was also high with all the conditions of Study 2 (median: 6/7) except for the 6 km/h condition, which was rated with a median of 5/7.

Task Difficulty. Participants found the reading task with the ◯ circular trajectory easier (median: 3/7) than the two other conditions (median: 3.5/7 and 4/7). In Study 2, participants found the reading task easy (median: 2/7) but slightly more challenging with the 6 km/h condition (median: 3/7). The task difficulty was rated as very easy while sitting and standing (median: 1/7).

Subjective Mental and Physical Load. In Study 1, participants found the task more mentally (median: 3.5/7 and 4/7) than physically demanding, with a median of 1/7 for the ⟷ line trajectory and a median of 2/7 for the ◯ circular and the ∞ infinity trajectories. In contrast, in Study 2, participants did not find the task mentally demanding (median: 2/7). However, the 6 km/h condition was rated more physically demanding (median: 3/7) than the other conditions, with a median of 2/7 for the 4 km/h condition and a median of 1/7 for the other conditions.

V. Discussion

This paper addresses challenges associated with reading micro visualizations in motion, focusing on scenarios in which both the viewer and the micro visualization itself are moving. We measured the characteristics of smartwatch movement when wearers are in motion and observed effects of this motion on reading performance. Next, we reflect on the key findings from our two studies and the challenges we encountered in studying visualization in motion.

A. Result Discussion

In this section, we discuss our results and their potential implications in light of our four research questions.

*1) **[RQ1] The Influence of Motion on Reading Performance**:* The study measured two types of reading accuracy: finding the chart with the highest progress and reading the percentage value of this chart. Finding the right chart seemed simple for participants, with accuracies ranging between 93.5% (fastest speed in Study 2) and 98.6% (standing in Study 2). We saw a larger range of differences with the second task, for which accuracy was also much lower, ranging from 70.4%–74.7% (∞ infinity trajectory – straight ⟷ line) in Study 1 and 74.5%–83.8% (fastest speed 6 km/h – standing) in Study 2. We, therefore, saw that for the first task people could easily maintain enough postural stability and upper-limb steadiness to find the correct radial progress chart while walking. This was confirmed by a participant's comment *"Jogging, I am swinging my arms more [...but...] I could still easily hold my wrist steady and focus on the data. (P13-S2)"* While this participant and the overall results showed that jitter was not always too impactful, we still noticed that increased jitter during motion coincided with less accurate reading of exact values from the visualization. One participant mentioned, for example: *"The watch moved around much more while jogging, so it was harder to read exactly. (P1-S2)"*

We found weak evidence in the data that participants made a trade-off between reading accurately and responding quickly while walking at higher speeds, with participants taking less time to answer, the faster they walked. Perhaps participants experienced a sense of urgency to complete the reading quickly and return to a normal walking posture, prioritizing their physical stability. One participant mentioned *"I had some mistakes, but I also wanted to answer somewhat fast and didn't want to keep looking at the watch for too long. (P7-S2)"*

We conclude that errors in reading likely stem from multiple factors, including divided attention and gait-induced visual perturbation. However, according to our measurements (Table V, Row 2) and participants' feedback, jitter induced by arm and body movements during

walking and especially jogging seems to be the most prominent contributing factor, depending on the task. In the following, we summarize our considerations for reducing this jitter as a direct implication of these findings.

Consideration: Reducing the jitter effect. Using screen-stabilizers, such as the SteadyScreen service,[2] which smartwatches support, could minimize the effect of jitter caused by the body oscillation while moving. This might already be useful at the moderate walking and slow jogging speed we tested, but it remains to be seen how effectively people can read micro visualizations in even more high-paced sports, or whether they can afford to do so at all; in which case, feedback would need to be given through other modalities, such as speech or screens attached to sports equipment, like handlebars.

Consideration: Visualization redesign for performance. Another possibility to combat the problem of jitter, is to redesign micro visualizations to improve how correctly people can read them. Due to limited display space, micro visualizations usually show only few or no reference structures (e. g., data axes and labels). This makes finding and reading the displayed information more difficult, potentially increasing the likelihood of errors. Finding ways to add readable reference structures, even in small display spaces, might help people see details faster. Examples include larger tick marks for important points in the data, discretizing the progress chart, and dual-encoding using color or text with increased font size.

*2) **[RQ2] The Influence of Motion on Reading Posture:*** We found several differences but also similarities in reading postures between our conditions and even the two studies. For example, people held their smartwatch at an average of around 28–31 cm from their eyes in both studies. This is a similar range observed in prior work [5]. With this measurement in place, we can more accurately judge the size, in degrees of visual angle, at which participants see micro visualizations on a smartwatch. Theoretically, holding a smartphone or smartwatch closer to one's eyes has a similar effect as increasing the size of the displayed information. Surprisingly, in our study, people held their smartwatch slightly closer when sitting and not walking. One participant mentioned that they could afford to do so only in the conditions without motion: "*It is easier to hold the smartwatch closer to my face, and it does not jiggle at all, so it's pretty easy to read.* (P15-S2)" This sentiment is understandable because, during motion, it is beneficial to hold the arm farther away to prevent the watch from obstructing the field of view and to reduce the perceived motion of the visualizations.

As visualization designers, we cannot control how people hold their watches, but we can control how visualizations are rendered. Our observations suggest that micro visualization type, size, information density, and font size should be studied to support readability across mobility conditions, in which people tend to hold their watches at varying distances from their eyes. Next, we summarize our suggestions for improving readability under these changing viewing distances.

Consideration: Adjusting micro visualizations depending on the activity. Prior work on smartphones [15] and in-vehicle displays [52] found that increasing the size of letters or icons helped to combat the negative influence of walking speed on visual acuity. Tan et al. [52] showed that increasing characters' size ($>$ 20 pt) improved visual search speed and accuracy in a vibration-prone automotive environment. We can assume that larger visualization size might similarly improve reading micro visualizations on smartwatches. Given our finding that people held their watches further away when walking faster, visualizations could be increased in size when the watch detects a higher running speed (see Fig. 7) or specific postures based on its internal sensors [36]. In addition to increasing the size of micro visualizations, the distance between the worn smartwatch and the viewer could be used for semantic zooming. For instance, when the smartwatch is brought closer to the viewer's eyes, more detail could be shown, including highlights or labels. Additionally, when the Alpha-Floor angle is small or converges to 0°—meaning that the watch is turned away from the viewer—the screen brightness is affected [6]. Therefore, dynamically increasing the screen brightness and switching to a brighter color scheme, similar to the gyro effects[3] example by the Samsung Watch, would help enhance readability.

Our measurements on distances and smartwatch angles can further help to run controlled confirmatory studies on the effect of dynamically changing visualizations. In these studies, an average distance of 28–31 cm from the screen could be used, and a rotation of the content with an Alpha-Floor angle of around 27° could simulate an average watch orientation. Yet, we observed larger differences in the Beta-Floor and Gamma-Head angles between the two studies, which indicates that the type of walking or navigation plays a role. More studies are needed to quantify the influence of these angles on reading performance, as prior work indicates that performance implications may vary across viewing angles [6].

*3) **[RQ3] The Influence of Motion on Task Load:*** Between our two studies we saw a difference of ≈2 s for average response times. One possible reason for this difference is the coordination between reading and navigating that participants had to do in Study 1. Participants commented on the navigation and reading challenges: *"It was easier to read data while walking in long continuous circles than short straight lines. (P1-S1)"* Another participant mentioned: *"There were less obstacles and I knew I had to walk in the same circular trajectory [...] it took less effort to read the data and compare and answer. (P9-S1)"* This further confirms that the more discontinued and filled

[2]https://github.com/Sublimis/SteadyScreen

[3]https://developer.samsung.com/codelab/watch-face-studio/gyro-effects.html



Fig. 7: Illustrations showing (a) our watch face stimulus, (b) how it might look if we increase the size of all the visual items uniformly versus (c) when increasing the size of a specific item based on the viewer focus.

with obstacles a walking path is, the harder reading micro visualizations on a smartwatch can be. This aligns with the higher mental load participants reported in Study 1 compared to Study 2. One participant mentioned this when reporting about the infinity trajectory: *"The task is not really more challenging or difficult but a little more mentally demanding [...] I'm also thinking about where I will be going at the same time. (P5-S1)"* This confirms again that synchronizing the reading and navigation tasks in the first study required more attention from participants. In contrast, none of the walking conditions in both studies were rated as physically demanding.

Consideration: Making smartwatch visualizations glanceable Our findings suggest the importance of making micro visualizations glanceable within a few hundred milliseconds to reduce mental load. One way to achieve this is to focus on *pre-attentive visual channels* in the data representation [12], for example, to visually highlight the relevant information. Such encodings can minimize the time taken to read micro visualizations and possibly help reduce the mental load that multitasking induces. In this direction we conducted a first investigation with three pre-attentive variables (i. e., area, color, and flicker) to assess their effectiveness in a target identification task with four micro visualizations. The study findings showed that color was the most effective and the most motion-robust [21]. However, further research is necessary to cover additional pre-attentive variables and reading tasks on smartwatches to test their effectiveness when informing the design of glanceable and in-motion visualizations.

*4) **[RQ4] The Influence of Reading Micro Visualizations on Walking Speed**:* When performing two tasks simultaneously (walking and reading), both will influence each other. Therefore, we also discuss the task's effect on walking speed in Study 1 (speed was fixed in Study 2). We observed that participants reduced their walking speed to focus on the task, especially with the infinity trajectory. This echoes findings from prior work, which showed that people slowed down when reading on smartphones while walking. Rubio Barañano et al. [47] hypothesized that this happens because during slower and smoother motion, it is easier for people to couple head and phone motion perceptually, while Krasovsky et al. [33] name visual task requirements, task-dependent cognitive requirements, and fine motor skills as possible causes. As such, it is plausible to consider movement speed, together with reading accuracy and time, as an indication of the difficulty of the task. We, therefore, recommend including the measurement of movement speed in future studies.

In addition to a discussion of the results we obtained, we now reflect on lessons learned about running micro visualizations studies on smartwatches while in motion.

B. Studying Smartwatches in Motion

*1) **Measuring reading performance with participants in motion**:* In our two studies, participants were wearing the smartwatch while walking overground (Study 1) and on a treadmill (Study 2). In both cases, it was difficult to measure reading performance because we had to carefully, and partly manually, track and calculate when the watch was actually being read. Was this effort worth it? There is evidence that the swinging movement of the arms while walking is essential for human dynamic stability [8], and maintaining one arm bent and relatively stable for more than a few seconds can alter the normal walking posture and might lead to the risk of falling. As such, studying the trade-off of how people hold their arm and read their smartwatch is important to capture the effectiveness of micro visualizations in different scenarios. Moreover, the arms are not the only parts of the body that motion affects; the viewer's head and their eyes are constantly moving as well. The good news is that our visual system, via the Vestibulo-Ocular Reflex (VOR), is capable of accommodating head motion by stabilizing the retinal image, ensuring that the LOS remains fixed on a target [19]. In an attempt to modulate the phone-to-eyes relative motion and investigate its influence on improving reading performance, Rubio Barañano et al. [47] asked participants walking on a treadmill to wear an elbow brace, restricting the motion-coupling between the phone and the eyes. Their results showed that the reading performance decreased when the arm was stabilized with an elbow brace and when the smartphone was fixed on the treadmill, than when holding the smartphone with a freely swinging arm. This indicates that walking participants should indeed be wearing their smartwatches for more realistic conditions.

*2) **Treadmill vs. overground walking**:* Studying visualizations in motion on treadmills rather than having people walk on the ground has several advantages. Motion tracking becomes much simpler, walking and running speed can be controlled, treadmills do not require a large space, and they can be equipped with further sensors, screens, or input devices. On the other hand, while treadmill and overground walking exhibit numerous similarities, they differ significantly in terms of energy expenditure and biomechanics, for example, pelvic range of motion and variations in joint moments and powers [50]. Prior studies [54] showed that people walk on the treadmill with

shorter stride lengths and higher cadence. Additionally, walking on the treadmill lowers vertical ground reaction force at push-off and joint moments at the knee and ankle. All these differences suggest that treadmill walking may alter natural gait mechanics, potentially affecting the vertical oscillation less and inducing less jitter. This might explain some of our differences related to viewing angles or even performance, for which participants' answers were 6% more accurate when reading the proportion percentage while walking in Study 2 than in Study 1. In summary, we found several similar results on the treadmill (LOS distance, Alpha-Floor angle, and correctness of the overview task) to overground walking, but the additional jitter and navigation task also created differences in response time, certain angles, and correctness on the detailed reading. Researchers need to balance these trade-offs by carefully considering what they care to measure and possibly consider treadmill conditions to produce "best case" results.

3) ***Treadmill settings****:* Prior studies [15] inspired the three walking speeds we used in Study 2, which we further calibrated to align with the average overground walking speeds from Study 1. During the study, participants provided spontaneous feedback on the treadmill walking speeds. Some participants described the 2 km/h condition as extremely slow. On the other hand, participants' opinions on the 6 km/h condition were more controversial; one group found it not sufficiently fast to be acceptable, while others were exhausted when completing the condition. Individual activity levels (e. g., fit versus non-fit) and physiological factors (e. g., step length, which correlates with height) appear to primarily influence these differences in preferences. Future research could explore walking at participants' preferred speeds or focus on recruiting regularly active individuals to evaluate performance at higher walking speeds (i. e., > 6 km/h).

4) ***Going beyond the present studies****:* Having measurements about viewing distances and angles can also help to position or simulate smartwatches for more controlled experiments and perhaps even to recreate relative motion using actuated stands to test the influence of coupling between hand and smartwatch motion and, ultimately, the demands on the oculomotor system. Such studies could help to explain differences between past and future studies and potentially help to explain results. For instance, scenarios might be studied in the future, in which a viewer needs to maintain their posture, for example, while cycling [34], at the same time, they have to keep an eye on their performance analysis (e. g., fitness performance and target trajectory to follow). Comparing the reading of micro visualizations in forced postures to free ones can help to explain differences in posture further.

VI. Conclusion

Via our two studies, we examined postural measurements when reading micro visualizations on a smartwatch and the influence of two motion characteristics: walking trajectory and walking speed. Reading micro visualizations on smartwatches can be a challenging task, especially when reading details (e. g., proportion estimation). We found that performing the reading task in motion created additional challenges, such as an increased perceived mental load when coordinated with a navigation task or the instability of the smartwatch due to body movement from high-speed motion. Our results showed that walking negatively impacted reading performance in general. However, people could read data relatively accurately while walking different trajectories and speeds of up to 6 km/h. This is good news, because it means that fitness data on smartwatches, similar to ours, can be read effectively on the go. Our results also suggest that the walking activity affected participants' reading postures. Finally, our work points to several concrete implications for smartwatch design but also future research. In both cases, our measurements and results can help to inform both micro visualization and study design as well as data analysis. In our own future work, we are particularly interested in examining and empirically assessing the effectiveness of our design implications for micro visualizations in realistic mobile reading contexts.

Acknowledgments

This project was funded by the DFG grant DFG-ER272-14, DFG under Germany's Excellence Strategy (EXC 2075–390740016). Fairouz Grioui is funded by the Alexander von Humboldt Foundation, funded by the German Federal Ministry of Education and Research. Lijie Yao is funded by the National Natural Science Foundation of China (NSFC), grant № 62602522. Petra Isenberg is funded by ANR-24-CE33-4945 SPORTSVIZ. Tanja Blascheck is funded by the European Social Fund and the Ministry of Science, Research and Arts Baden-Württemberg.

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